Applying Behavioral Entity Intelligence to Forecasting, Supplier Risk, and Mission Readiness

Decision Intelligence for Defense Logistics

Innovative Concept Paper for DLA

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1. Executive Summary

In plain terms, EDM / BEI models every part, supplier, depot, and route as a living profile and watches which way its behavior is heading, not just where it stands today, so DLA can act before a threshold breaks. On worked demand items it cut the 8-week buy by 40 to 54 percent (about $460,000 less over-buy on one part) while still covering realized demand, and it produced forecasts for items that legacy methods report as zero. The results below are a proof of concept on synthetic data and would be re-validated on DLA data.

Defense Logistics Agency (DLA) operates in a contested logistics environment where adversarial exploitation and disruption can emerge through demand volatility, supplier fragility, sub-tier dependency, counterfeit exposure, sourcing shifts, route vulnerability, cyber-enabled disruption, or correlated movement across the defense industrial base. In this environment, the mission risk is not simply a late shipment, an inaccurate forecast, or a delayed supplier alert. The mission risk is the loss of decision time, the interval between the first observable behavioral signal and the point at which readiness, mission assurance, or operational resiliency is affected.

As the Nation's Logistics Combat Support Agency, DLA must sustain warfighter readiness across a global supply chain that is increasingly nonlinear, relationship-dependent, and exposed to disruption. In a contested logistics environment, adapting faster, resolving supply-chain friction earlier, and preserving decision time are essential to mission assurance, operational resiliency, and U.S. military superiority. Current planning, forecasting, and supplier-monitoring methods remain necessary, but they can leave DLA reacting after thresholds break rather than seeing where demand, supplier behavior, inventory posture, or network exposure is moving. The requirement is a stronger decision-intelligence layer that improves supply-chain risk mitigation and supports decision advantage before operational consequences are fully visible.

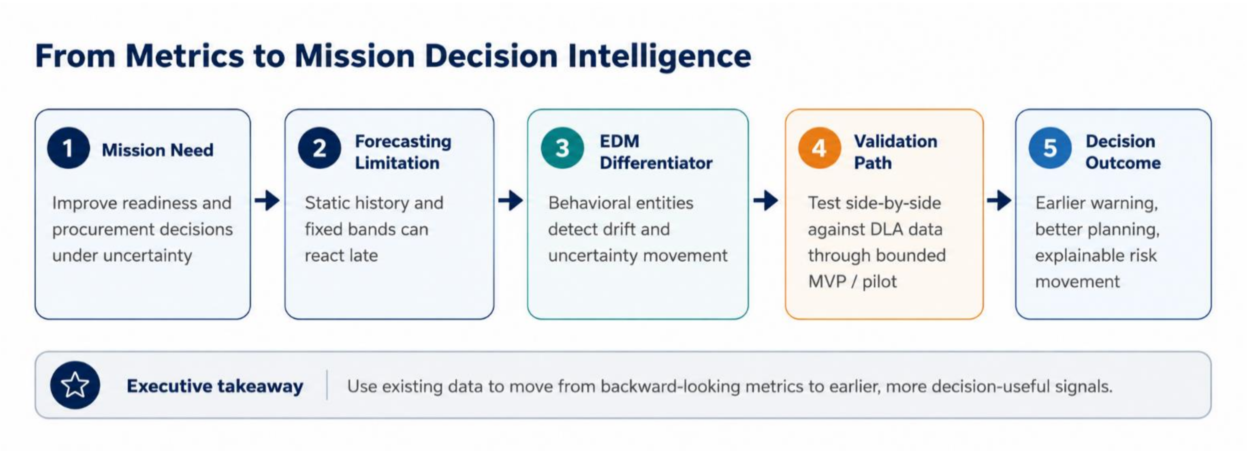

Figure 1 - From Metrics to Mission Decision Intelligence

This paper presents Entity Digital Model (EDM) Vector Intelligence and Behavioral Entity Intelligence (BEI), an innovative decision-intelligence approach that brings the representational power behind large language models and transformer architectures down to the application layer of defense logistics. The core innovation is a new way to represent the problem: EDM / BEI models logistics data as connected, time-aware digital twins.Each item-location, supplier, depot, route, relationship, and network is represented as a behavioral entity. Its movement is measured over time and compared against its own history and peer cohorts. It is aligned to known supply-chain concepts and translated into explainable planning or risk signals. Applied across demand, supply, and multiple logistics scenarios, it right-sizes procurement where conventional methods over-order and catches demand they miss entirely. It also provides capabilities legacy methods cannot, such as calibrated uncertainty, early warning from behavioral drift, and forecasts for items with little or no history.

The recommended next step is a bounded pilot that runs current DLA baselines and EDM / BEI outputs side by side across selected NSNs, suppliers, depots, and mission-relevant scenarios. Use Case 1, Demand Forecasting and Inventory Planning, is the prioritized starting case, because it provides the clearest path to validate forecast quality, uncertainty management, procurement-plan accuracy, early warning, and planner usability against existing DLA data. From there, the same operating model extends to supplier-risk and logistics-network early-warning use cases in support of broader mission assurance, operational resiliency, supply-chain risk mitigation, and decision advantage.

The detailed proof-of-concept results are summarized in Section 2, “Proof of Concept: Decision-Value Evidence,” which explains how the current proof of concept supports the recommendation for a bounded pilot using DLA operational data.

2. Proof of Concept: Decision-Value Evidence

The current proof of concept demonstrates that EDM/BEI can produce decision-relevant outputs across DLA-aligned demand forecasting, supplier risk, and network early-warning scenarios. The results should be read as proof of concept, not as operational proof on DLA data. They show that the model architecture can generate calibrated planning ranges, supplier-risk discrimination, relationship-level visibility, explainable drivers, and early warning signals, warranting validation through a bounded pilot using DLA operational data.

EDM turns forecasting into planning intelligence. It models each item-location as a behavioral entity and uses demand, supply, inventory, volatility, and context signals to adjust the P10 to P90 planning interval: it reads 16 to 23 monthly snapshots, measures behavioral drift, aligns that drift to supply-chain concepts, and produces calibrated planning ranges for procurement and early-warning decisions. Because this proof-of-concept window spans roughly two years, it covers only about two annual cycles; full seasonal and exercise-cycle validation requires the longer DLA demand history and is a pilot data-readiness item.

The core innovation is not another forecast but a new way of representing the problem. Each item and depot is rebuilt as a digital twin using EDM Vector Intelligence, with its state embedded in a high-dimensional vector space. Raw requisition, supply, and inventory records become a living, behavioral entity rather than a flat table. That representation is what makes advanced operations possible, including behavioral drift detection, concept alignment, peer-cohort transfer, and full distributional simulation, techniques that generic tabular data cannot support and that let the approach answer questions a conventional forecaster cannot.

On three representative items, EDM Vector Intelligence substantially reduced over-procurement against the moving-average baseline. For an adapter plate it cut the 8-week P90 planning buy from 379 to 225 units, roughly 40 percent lower and about $460,000 less over-procurement; for a needle bearing it cut the buy from 1,184 to 545 units, roughly 54 percent lower and about $48,000 less. It also corrected the opposite failure: for a valve train kit with no recent history, the baseline forecast zero while EDM inferred 95 units from similar items, catching demand that legacy methods miss entirely. These are per-item results on synthetic data and would be re-validated on DLA data.

EDM Turns Forecasting into Planning Intelligence

EDM models each item-location as a behavioral entity and uses demand, supply, inventory, volatility, and context signals to adjust the P10 to P90 planning interval.

- Reads 16 to 23 monthly snapshots and measures behavioral drift
- Aligns drift to supply-chain concepts and produces calibrated planning ranges for procurement and early warning
- Adapter plate: 8-week P90 buy cut from 379 to 225 units, about $460,000 less over-procurement, while still covering realized demand
- Needle bearing: 1,184 to 545 units, about $48,000 less; valve train kit: 0 to 95 units, catching demand legacy reports as zero
- Worked examples on synthetic data; to be re-validated on DLA data

Those distributions are trustworthy because the model's probabilities are tightly calibrated. On the occurrence stage, which predicts whether demand will appear at all, it reaches an Expected Calibration Error (ECE, the gap between predicted probabilities and observed frequencies, where lower is better) of 0.0322 and a Brier score (the mean squared error of a probabilistic prediction, where lower is better and zero is perfect) of 0.0885. Calibration at this level is what lets a planner act on the full range rather than a single estimate, and it is the foundation of the procurement and early-warning decisions the twin is built to support. These results were established on synthetic data and are ready for validation on DLA operational data.

BEI converts supplier risk into behavioral intelligence. It models supplier risk across connected behavioral entities, including suppliers, NSNs, locations, routes, manufacturers, countries of origin, contracts, and sub-tier relationships, and surfaces interpretable drivers such as quality incidents, recent order volume, on-time rate, worst delay, and critical-NSN count. Modeling these as connected entities lets the approach see how supplier behavior and relationships are moving, rather than only whether a current-period metric has crossed a threshold.

The supplier-risk approach moves beyond static supplier scoring by evaluating how risk develops through supply-chain behavior and relationship patterns. Rather than treating risk as a single supplier attribute, BEI analyzes how operational performance, sourcing exposure, delays, and dependency patterns combine to affect mission readiness. Alongside a point-in-time risk view, the survival model estimates how risk develops over 30, 60, and 90-day horizons, so action can be taken before the window closes. The logic is built to be interpretable: monotonic constraints ensure that stronger on-time performance lowers estimated risk while quality incidents, sole-source exposure, and severe delays raise it, and explainable drivers show why a supplier is flagged. Applied as a full behavioral model, the same representation extends to relationship and sub-tier exposure and to drift that surfaces before current-period metrics change, which is where the approach adds the most for supplier risk.

BEI Converts Supplier Risk into Behavioral Intelligence

BEI models supplier risk across connected behavioral entities, including suppliers, NSNs, locations, routes, manufacturers, countries of origin, contracts, and sub-tier relationships.

- Surfaces interpretable drivers: quality incidents, recent order volume, on-time rate, worst delay, and critical-NSN count
- Estimates how risk develops over 30, 60, and 90-day horizons, alongside a point-in-time view

- Monotonic constraints keep the logic interpretable: stronger on-time performance lowers risk; quality incidents, sole-source exposure, and severe delays raise it
- Presented as a qualitative proof of concept requiring real-data validation

BEI detects mission risk before thresholds are breached. It provides a cross-domain early-warning framework that looks beyond static thresholds by analyzing entities, relationships, and networks over time. Each entity is described along nine behavioral dimensions (identity, behavior, time, space, relationship, network, sequence, periodicity, and context), and the framework applies three analytical layers: entity profiles, pairwise relationship dynamics, and graph-level network analysis. In a synthetic cybersecurity proof of concept it identified all four embedded long-duration attack campaigns at a false-positive rate of 8.1%, where conventional anomaly methods caught at most one. This is relevant to DLA because supply-chain and readiness risks may remain within nominal thresholds while their behavioral direction changes.

This early-warning approach surfaces weak signals that conventional monitoring may miss when activity still appears normal on the surface. The proof of concept used synthetic telemetry covering 250 users, 485 days, and approximately 14 million events, providing a large test environment for evaluating long-duration behavioral change. The benchmark showed stronger campaign detection than conventional anomaly methods: the z-score method identified one campaign, while the local outlier factor, Isolation Forest, and the one-class support vector machine identified none. The results should be read with appropriate caution: two of the four campaigns were identified marginally, at ranks 7 and 24 of 250, and the operating point was selected with knowledge of the ground truth, so the result is best positioned as a reporting benchmark rather than a deployable operational threshold. For DLA, the takeaway is that risk may develop gradually before it triggers a standard rule, metric, or threshold, and BEI helps identify changing patterns early enough to support investigation, prioritization, and readiness-focused response.

BEI Detects Mission Risk Before Thresholds Are Breached

BEI provides a cross-domain early-warning framework that analyzes entities, relationships, and networks over time.

- Describes each entity along nine behavioral dimensions and applies three analytical layers (entity, relationship, network)
- In a synthetic cybersecurity proof of concept, identified all four embedded long-duration attack campaigns at an 8.1% false-positive rate, where conventional anomaly methods caught at most one
- Offered as a transferable detection principle, that slow, persistent, sub-threshold signals are discoverable, and a hypothesis to validate on DLA data; the 8.1% operating point was ground-truth-selected, a reporting benchmark not a deployable threshold
- Relevant to DLA because supply-chain and readiness risks may stay within nominal thresholds while their behavioral direction changes

3. The DLA Decision Problem

DLA operates in an environment in which decisions must often be made before the relevant uncertainty is resolved. DLA's operations are vast and wide-ranging. Forecasting, supplier monitoring, inventory planning, and readiness support are just a few of its many key activities. In conventional analytics these are usually approached as separate problems, each with its own model and data, even though they are operationally coupled: a demand signal may rise because of mission activity, while the associated risk is governed by supplier capacity, depot position, lead time, part criticality, and network dependency. A supplier may appear compliant in aggregate while a single supplier-part-country relationship deteriorates, and a network may appear healthy in individual metrics while several nominally independent suppliers share a common fragile sub-tier source.

The recurring limitation is timing. Conventional methods tend to detect a consequence after the underlying behavior has already changed: a moving average responds once demand history shifts, a supplier score responds once delivery, quality, or financial thresholds deteriorate, and a network review responds once dependencies become visible through disruption. These methods remain necessary but are not sufficient in mission environments, where latent change can translate into procurement, readiness, or mission risk before a threshold-based alert is triggered.

The problem is therefore broader than model accuracy. DLA requires earlier and explainable signals that inform specific questions: what should be stocked, where risk should be buffered, which suppliers are drifting, which relationships are changing, which latent dependencies are material, and what should be validated before a larger operational commitment is made. The appropriate analytical target is not an additional dashboard indicator but a layer that characterizes where demand, supplier behavior, inventory exposure, and network risk are moving.

3.1 Why Traditional Analytics Fall Short

- **Threshold insensitivity: rules and supplier scores can stay green while sourcing, relationship, or network risk changes underneath them. (e.g., a supplier keeps meeting its on-time target so its scorecard stays green, even as its lead times creep up and it shifts to a higher-risk country of origin)**
- **Magnitude-only measurement: conventional anomaly models measure how much a metric changes, not what direction the entity is becoming. (e.g., a supplier's total order volume holds steady while its mix shifts toward critical, sole-source parts; the quantity is unchanged but the risk direction has moved)**
- **Single-entity scope: a supplier, NSN, depot, or user may look normal individually, while the relationship between entities has changed. (e.g., two suppliers each look healthy on their own, yet both depend on the same sub-tier manufacturer, so one disruption takes out both)**
- **Static uncertainty: static planning bands cannot narrow when demand calms or widen when volatility and disruption risk increase. (e.g., a fixed band holds a wide buffer through months of calm demand, then fails to widen fast enough when a mission surge hits)**
- **Delayed response: backward-looking methods react after the data changes, leaving less time to buffer inventory, investigate supplier drift, or reduce mission exposure. (e.g., a forecast reacts only after consumption has already spiked, leaving little time to position inventory or investigate the supplier)**

These limitations do not indicate that current methods lack value; they indicate where current methods should be complemented. The proposition advanced here is that DLA can retain transparent baselines while adding a behavioral layer that measures drift, characterizes its direction, and supports controlled validation before procurement or readiness consequences become observable.

4. Technical Foundation - Behavioral Entities, Drift, and Decision Signals

Behavioral Entity Intelligence, as shown in Figure 2, represents an observable subject as a profile that evolves over time rather than as a static record. An entity may be an item-location, supplier, NSN, depot, country-of-origin relationship, user, device, route, manufacturer, or network cluster. The representation captures the entity's attributes, its activity, its change over time, its spatial setting, its relationships to other entities, its sequential and periodic behavior, and the context that gives the behavior meaning.

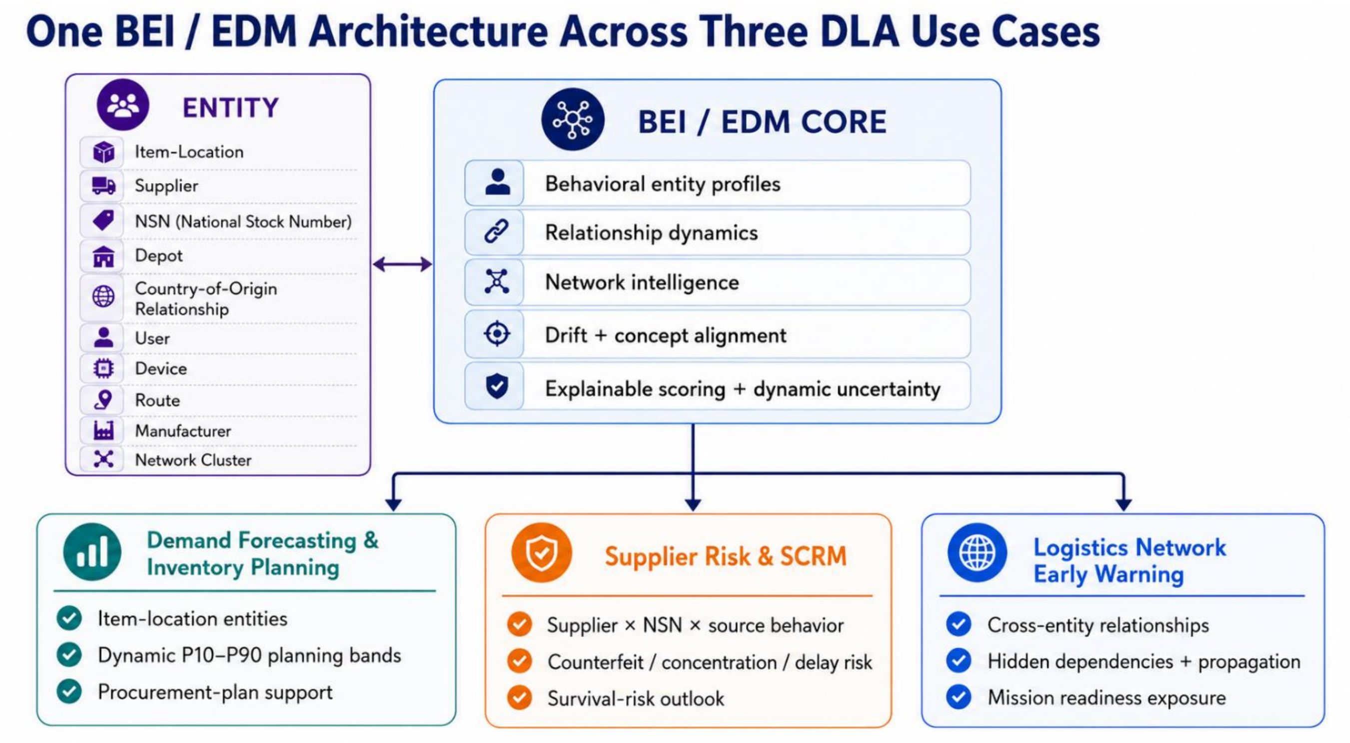

Figure 2 - One BEI / EDM Architecture Across Three DLA Use Cases

The central premise is that consequential risk frequently manifests as a change in what an entity does rather than in how much it does. A supplier may deliver a constant volume while shifting its sourcing toward a higher-risk geography; an item may exhibit sparse historical demand while its behavioral trajectory moves toward a disruption; and a network may present normal individual nodes while several relationships drift in a common direction. Conventional metrics quantify magnitude; BEI additionally characterizes the direction of behavioral change.

4.1 Key Behavioral Dimensions

In the current proof of concept, each entity is described along nine key behavioral dimensions: identity, behavior, time, space, relationship, network, sequence, periodicity, and context. These are the dimensions that matter most for the use cases shown here, not a fixed or complete set. Which dimensions apply, and how many, depends on the business problem, the operational scope, and the context available in the data, and the framework adds more as those grow.

Dimension	What It Captures	DLA Relevance
Identity	Relatively stable attributes of the entity	NSN, supplier, Commercial and Government Entity (CAGE) / Unique Entity Identifier (UEI), depot, platform, criticality, commodity, contract, tier

Dimension	What It Captures	DLA Relevance
Behavior	What the entity does across activity zones	Demand, delivery, quality, sourcing, inventory, financial, route, or access behavior
Time	Evolution, velocity, acceleration, and drift	Early warning when behavior changes before thresholds move
Space	Geography and topology	Depot position, country of origin, supplier location, theater, route, operational zone
Relationship	Pairwise interaction pattern	Supplier x NSN, Supplier x Country, NSN x Location, Supplier x Supplier
Network	Graph-level ecosystem intelligence	Hidden dependencies, common sub-tier exposure, correlated supplier movement
Sequence	Order of operations	Sourcing sequence, procurement event pattern, disruption sequence, maintenance path
Periodicity	Rhythm and cadence	Seasonal demand, exercise cycles, delivery cadence, and inspection pattern
Context	Environmental meaning	Mission posture, exercise activity, platform lifecycle, regulatory or geopolitical conditions.

Table 1 - Nine behavioral dimensions and their DLA relevance.

Context captures the situation an item sits in, not its own activity. In the current approach, it comes from the item's structured attributes, its criticality, category and supply class, the platform it supports and that platform's lifecycle stage, and whether it is exercise sensitive, which are written into a short sentence and embedded alongside the other signals. For example, a demand rise on a critical, exercise sensitive part on an aging platform means something very different from the same rise on a routine item, and context is what lets the model tell them apart. Richer context such as regulatory or geopolitical conditions would be added from real operational data.

4.2 Our Layered Analytical Architecture

Our approach consists of a layered analytical architecture, as shown in Figure 3 below.

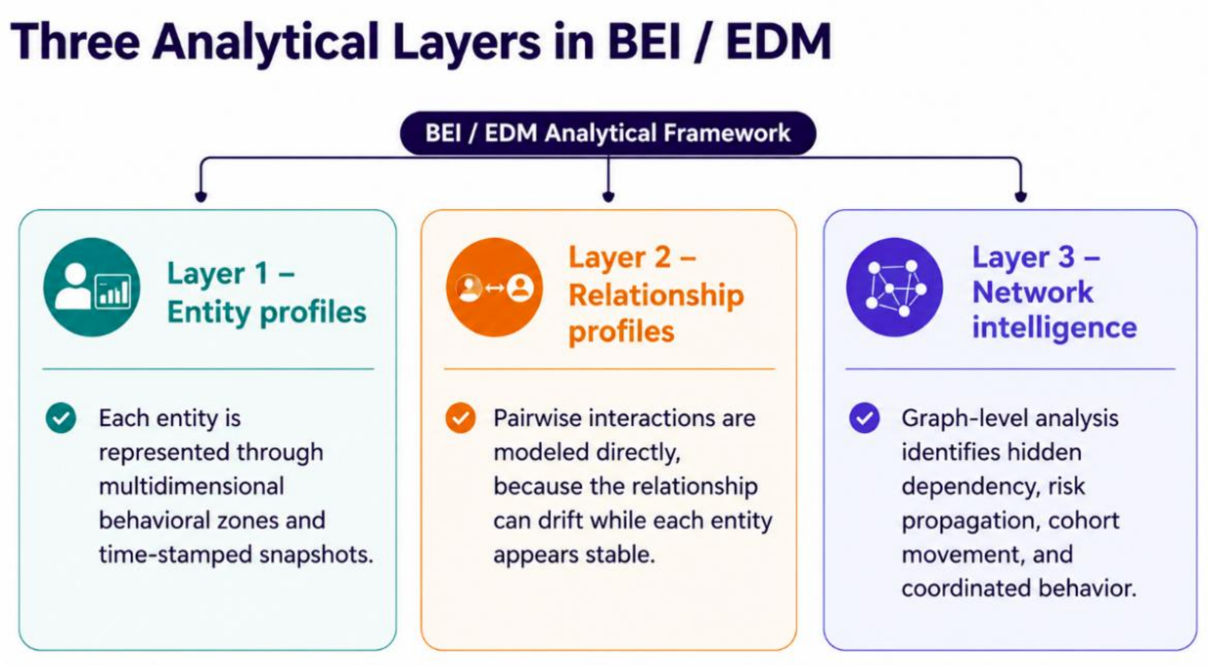

Figure 3 - Our Layered Analytical Architecture

What makes the architecture useful is the work each layer does. The first layer profiles each entity separately, whether a supplier, an item, a depot, or a route. Most analytics stop at this level. In practice, a large share of supply chain and security risks does not surface when entities are examined one at a time. The second layer treats the relationship between two entities as something to model in its own right, since a dependency can weaken even when both entities remain healthy on their own. Two nominally independent suppliers that begin to move together are a relationship signal, not an entity signal. The third layer steps back to the whole network, where risk propagation, hidden common dependencies, and coordinated behavior become apparent only in the graph. Taken together, the layers describe individual behavior, the links between entities, and the way risk spreads across the system.

Time is treated as a dimension that cuts across all three layers rather than as a fourth layer. Each entity, each relationship, and the network itself is stored as a series of time-stamped snapshots, and the model reads drift, velocity, and direction of change from that series. This is the part that separates the method from point-in-time reporting, because it lets the system act on where something is heading rather than only on its current state.

The architecture is best read as a direction of travel rather than a finished schema. Each layer is built to handle more data and attributes over time, including additional attributes at the entity layer, new relationship types at the relationship layer, and richer edges and propagation rules at the network layer. As the data foundation grows, the same three-layer structure can be widened and re-tuned without a redesign.

4.3 Mathematical and Model Design Principles

The architecture follows a consistent pattern. Entity behavior is serialized into structured text or feature descriptions, embedded in a high-dimensional vector space, compared against the entity's own history and its peer cohort, and scored based on drift, concept alignment, accumulation, relationship behavior, and predictive risk models. The contribution is the composition of these established techniques into an explainable decision system; no individual method, formula, or model is claimed to be novel in isolation.

Positioning relative to existing tools. EDM / BEI is intended to complement, not replace, the established demand-planning and supply-chain-risk markets. Demand-sensing and planning platforms (for example Kinaxis, o9, and Blue Yonder) and supplier-risk and n-tier visibility platforms (for example Interos, Everstream, and Resilinc) already provide control-tower visibility, probabilistic planning, and supplier monitoring. The differentiation claimed here is not any single technique but the composition: item-location behavioral drift, calibrated planning bands, peer-cohort transfer for sparse items, and a cross-domain early-warning method applied together at the application layer on DLA data. This positioning should be validated against incumbent tooling during the pilot.

Core operators include behavioral composites, cosine or distance-based drift, drift direction vectors, concept alignment, peer-cohort z-scores, cumulative sum (CUSUM)-style accumulation for slow drift, relationship profiles, and multi-phase composite scoring. In demand forecasting, those operators support dynamic uncertainty bands and procurement planning. In supplier risk, they support risk discrimination, survival timing, and SHAP-based explanation. In network early warning, they support zone-level directional intelligence and relationship/graph triage.

5. Use Case 1- Demand Forecasting and Inventory Planning

Defense logistics demand is intermittent and highly variable, frequently driven by exercises, deployments, platform lifecycles, maintenance cycles, and supplier behavior. Deployments in particular concentrate demand around operational hot zones, the theaters and regions where activity surges and specific parts are consumed far faster than baseline. The framework's space and context dimensions are built to read an entity's behavior relative to these zones, so a demand rise tied to a hot zone can be separated from ordinary variation. Richer deployment and theater data would sharpen this further.

Figure 4 - Demand Forecasting and Inventory Planning

As shown in Figure 5, the legacy demand approach remains important because it is transparent and fast. It examines one item at one depot, uses recent history, trend, exercise factors, and fiscal factors, then scales empirical quantiles through the same ratio. That makes it a valid baseline. Its structural limitation is the frozen band shape. Because P10 through P90 scale together, the P90 / P50 relationship does not adapt when demand calms, destabilizes, or shifts regime. Legacy also reacts backward, struggles with cold-start and sparse items, and cannot borrow behavior from similar items or related supply-chain entities.

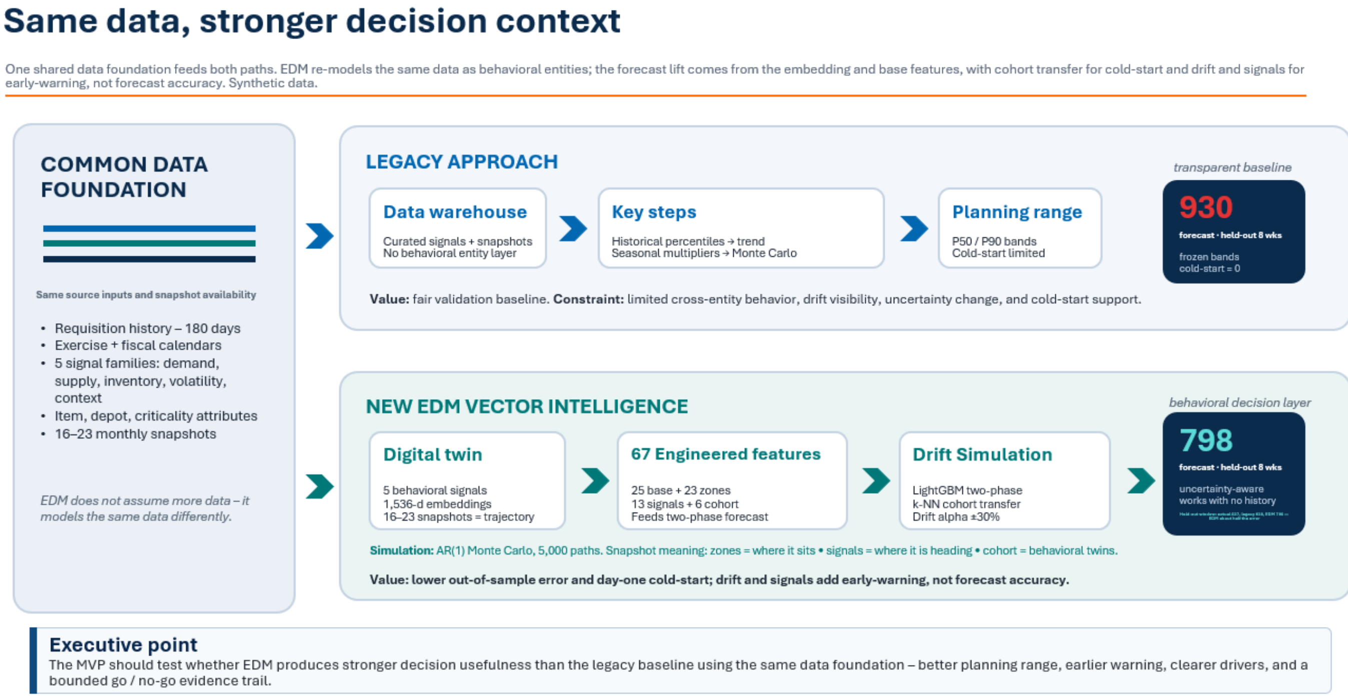

Figure 5 - Legacy Approach and EDM Vector Intelligence

EDM Vector Intelligence, shown in Figure 4, changes the unit of analysis. Each NSN x Department of Defense Activity Address Code (DODAAC) is modeled as a digital entity in a high-dimensional behavioral vector space. The entity is built from five behavioral signals: demand, supply, inventory, volatility, and context. Each month the entity is re-embedded, producing 16 to 23 monthly snapshots that form a trajectory. The model then asks two questions: what has this item been doing, and where is it heading?

5.1 EDM Forecasting Pipeline

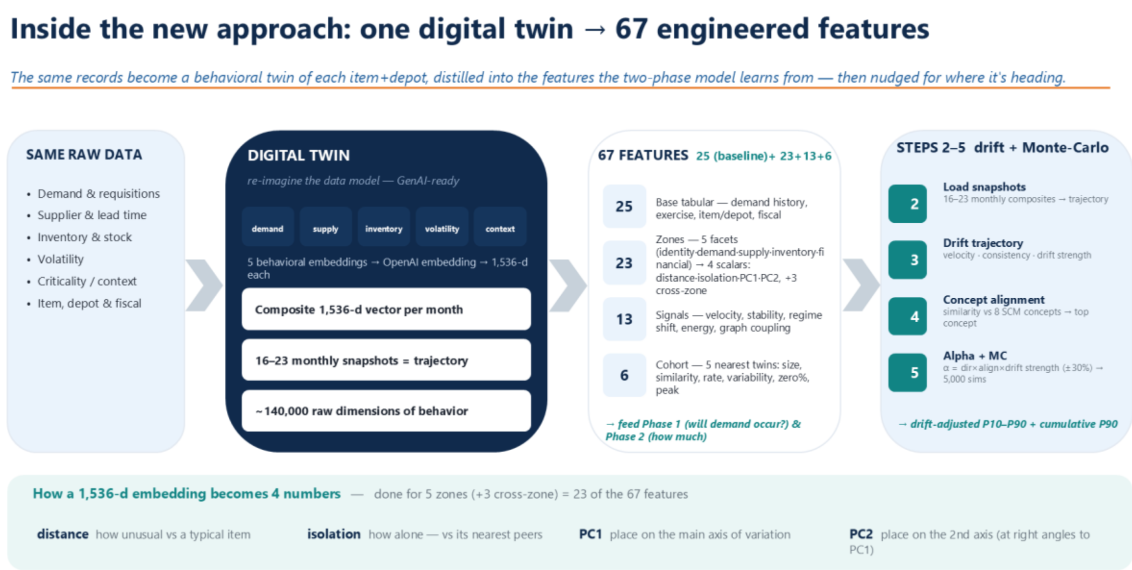

Figure 6 - Digital Twin Demonstration

- Digital twin: the same records are re-modeled as a behavioral twin of each item-and-depot as shown in Figure 6. Five signal families (demand, supply, inventory, volatility, and context) are serialized and embedded into 1,536-dimension vectors, composited into one monthly vector, with 16 to 23 monthly snapshots forming the trajectory.
- Engineered features: the twin is distilled into up to 67 features: 25 base tabular (demand history, exercise, item and depot, fiscal), 23 zone-derived (each behavioral zone reduced to its distance, isolation, and first two principal components, plus cross-zone terms), 13 drift signals, and 6 cohort features drawn from the nearest behavioral twins. The validated configuration used 29. Of the 67 features, 42 (the 23 zone-derived, 13 drift, and 6 cohort families) are generated from the embedding representation rather than from raw tabular fields; that is the embedding's specific contribution, and the tabular-only baseline (V1, 25 features) versus the embedding-enriched variant (V2, 29 features) isolates it.
- Cohort transfer: for items with little history, a nearest-behavioral-twins (k-NN) transfer produces a usable day-one forecast, addressing cold-start.
- Baseline: a two-stage LightGBM hurdle model separates occurrence from quantity. The first stage estimates whether demand occurs. The second stage estimates P10 through P90 quantiles and enforces monotonic quantile order.
- Snapshots: 16 to 23 monthly composite snapshots are loaded for each entity. If fewer than two exist, the system returns the baseline unchanged, which provides graceful degradation.
- Drift trajectory: Euclidean (L2) distances between consecutive snapshots generate velocity, acceleration, mean-step drift, consistency, drift intensity, and drift norm.
- Concept alignment: drift direction is projected against eight pre-embedded supply-chain concepts by cosine similarity, so the model can name what the entity is becoming.
- Alpha and dynamic bands: the model converts concept alignment into a bounded plan adjustment and a band multiplier, then reruns Monte Carlo to produce adjusted quantiles and 8-week safety-stock targets.

5.2 Demand Planning Decision-Value Evidence

The current proof-of-concept results show that EDM translates the same demand, supply, inventory, volatility, and context signals into calibrated planning ranges and right-sized procurement, reducing over-procurement on items the baseline over-orders and catching demand the baseline reports as zero, as quantified above. These are decision-value results on synthetic data, not an operational claim. The next step is to validate them on DLA operational data through a bounded pilot that compares EDM / BEI outputs against existing baselines and SME judgment.

The worked examples below show both failure directions on representative items (8-week P90 forecasts, synthetic data). These worked-example forecasts and the calibration figures reported below come from the validated embedding-enriched configuration (V2, 29 features); the 67-feature count is the full architecture, of which not all layers were active in these measurements.

Item (NSN)	Legacy 8-wk P90	EDM 8-wk P90	Outcome
Adapter plate (5120-01-397-8469)	379	225	40% smaller buy, about $460K less over-procurement
Needle bearing (3110-01-614-2738)	1,184	545	54% smaller buy, about $48K less over-procurement

Valve train kit (2910-01-905-6139)	0	95	catches demand legacy reports as zero, about $28K

Table 2 - Demand worked examples (synthetic data). Source: EDM Vector Intelligence vs moving-average baseline, 8-week P90.

Measured against realized demand on these items, EDM's planning bands track actual requisitions far more closely than the legacy bands. For the adapter plate, the EDM 8-week P90 of 225 units sits at the observed peak 8-week demand of 224 units, while the legacy P90 of 379 carried roughly 155 units of excess. For the needle bearing, the EDM P90 of 545 is far nearer realized demand than the legacy P90 of 1,184, which over-forecasts by several hundred units. In each case EDM right-sizes toward real demand rather than padding the band.

Because the proof-of-concept data carries an item criticality code (the adapter plate and needle bearing are coded C and the valve train kit B), the planning quantile can be tiered by criticality: a higher service level (for example P95 to P99) for sole-source, critical, or war-reserve items, and a lower level for routine items, rather than a uniform P90.

Note: *the National Stock Numbers shown in these examples are illustrative synthetic identifiers used for the proof of concept; they do not correspond to actual catalog records.*

Technical Point	Source Detail	Decision Interpretation
Synthetic scope	500 items, 200 suppliers, 4 depots, 24 months, approximately 277K requisitions	Large enough for concept testing, but still requires DLA-data remeasurement
Entity representation	high-dimensional vectors (current pilot: 1,536-d), five signals, 16-23 monthly composites	Forecasting becomes trajectory-aware rather than row-based
Feature strategy	25 base tabular features (V1); 29 in the measured embedding-enriched variant (V2); up to 67 with all EDM layers active	Technical specificity without overstating measured configuration
Calibration	Occurrence ECE 0.0322 and Brier score 0.0885, Grade A	Probability outputs are credible enough to test safety-stock planning
Quantile coverage	P10 0.141, P25 0.257, P50 0.498, P75 0.743, P90 0.894	Quantile coverage is close to nominal, with conservative lower-band behavior
Procurement example	Adapter Plate NSN 5120-01-397-8469 at W25H1R, legacy P90 379 vs. EDM P90 225	154 fewer P90 units, approximately $460,000 avoided over-procurement on one item
Disruption example	Sole-source critical item aligns with supply_disruption, alpha capped at +30%, bands widen 1.4x	Potential 1-3 month head start before disruption appears in delivery counts
Volatility example	Volatility_increase concept widens bands 1.6x with median unchanged	Model signals uncertainty without pretending to know direction

Technical Point	Source Detail	Decision Interpretation
Operational safeguards	Alpha capped at +/-30%, weak alignment zeroed, inconsistent drift damped, exercise-sensitive rising demand protected	Prevents unstable procurement movement from noisy drift

Table 3 - Demand forecasting technical evidence (synthetic data).

The demand use case should be evaluated on procurement-plan accuracy, calibration, pinball loss, mean absolute scaled error (MASE), horizon-level performance, early warning, and planner usefulness. Point-accuracy metrics should remain visible, but it should not become the only scorecard because the EDM value proposition is the distribution and its direction, which is a more useful basis for planning.

6. Use Case 2 - Supplier Risk and Supply Chain Risk Management (SCRM)

We understand Supply Chain Risk Management (SCRM) as a compliance-driven, mission assurance discipline that must operate across the full supplier, technology, logistics, and service lifecycle. Our approach aligns SCRM governance with federal acquisition, cybersecurity, and supply-chain assurance requirements, including NIST SP 800-53, NIST SP 800-161, DFARS, NDAA, FAR-based sourcing controls, and related federal risk-management expectations. It emphasizes supplier qualification, exclusion-list screening, contractual flow-downs, OEM and authorized-reseller sourcing, component authenticity, chain-of-custody documentation, serial-level traceability, anti-counterfeit inspection, cybersecurity validation, incident reporting, and continuous updates to the SCRM plan as supplier profiles, regulations, threat intelligence, or operational conditions change.

For DLA, this compliance foundation is directly relevant to supplier risk, mission assurance, and supply-chain risk mitigation in a contested logistics environment. Adversarial exploitation and disruption may emerge through counterfeit or prohibited components, hidden sub-tier exposure, sourcing shifts, cyber-enabled compromise, supplier concentration, delivery disruption, or regulatory noncompliance before conventional supplier metrics show failure. Our SCRM posture provides the governance and control foundation for identifying, assessing, mitigating, and reporting these risks. EDM / BEI builds on that foundation by modeling suppliers, NSNs, locations, routes, manufacturers, contracts, countries of origin, and sub-tier relationships as connected behavioral entities, enabling DLA to evaluate where supplier behavior and supply-chain exposure are moving before readiness or operational resiliency is affected. Supplier risk, as shown in Figure 7, is better treated as a continuous mission and readiness consideration than as a periodic compliance check, because it can develop while current-period supplier metrics remain acceptable. A supplier may continue to deliver within threshold, pass quality checks, and meet price expectations while changing its sourcing origins, sub-tier dependencies, manufacturer relationships, or concentration exposure. Conventional monitoring confirms observed failure and frequently does not detect such drift.

BEI for supplier risk applies the same entity principle to suppliers, NSNs, locations, routes, manufacturers, countries of origin, contracts, financial indicators, and sub-tier relationships. It asks what the supplier is becoming, how its relationships are changing, and whether those changes are unusual for its cohort. That matters because many supplier risks appear in relationships before they appear in aggregate supplier metrics.

6.1 Supplier Behavioral Zones and Relationships

Our Supplier Risk Zones and Relationships are detailed in Figure 7, below.

This is the three-layer Behavioral Entity Intelligence architecture applied to supplier risk. Rather than reduce a supplier to a single score, EDM Vector Intelligence represents each supplier as a behavioral entity described across several zones, and it treats the connections between suppliers, parts, and locations as objects in their own right. Figure 7 shows both. The top row is the entity layer: five behavioral zones, each tracking a different aspect of the supplier (identity, performance trajectory, geographic and sourcing exposure, network position, and financial standing) and the risk that surfaces when that zone drifts. The lower half is the relationship and network layer: it maps how suppliers, NSNs, countries of origin, depots, and bills of materials connect, where exposures such as hidden sub-tier dependency or adversary-nation sourcing become visible only when entities are viewed together.

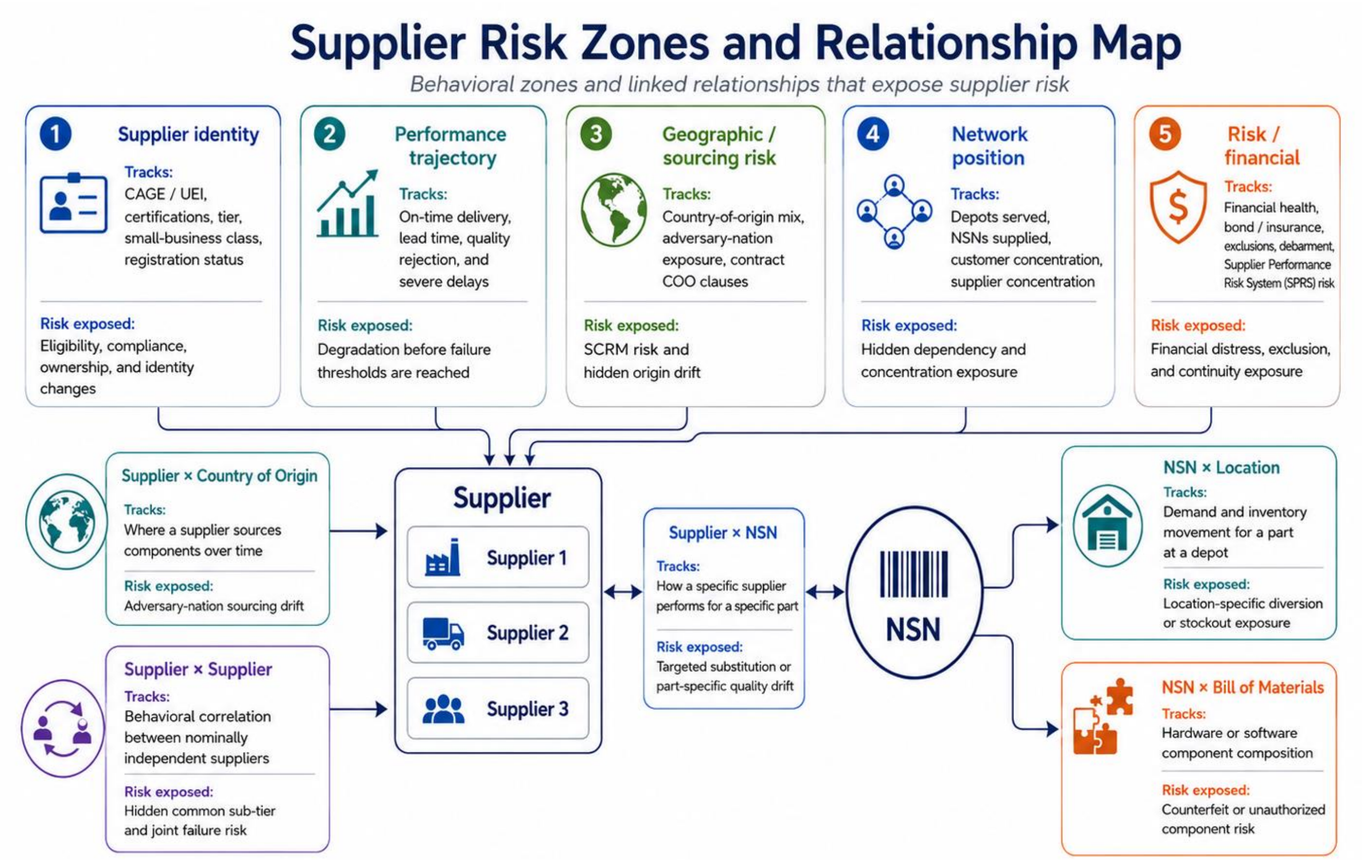

Figure 7 - Our Supplier Risk Zones and Relationship Map (Target Architecture)

6.2 Supplier-Risk Technical Proof Points

In a proof-of-concept modeling phase, a supplier-risk classifier and a survival model were trained on synthetic data. The early results show that the behavioral approach transfers from demand to supplier risk: the models flag higher-risk suppliers and estimate time-to-failure with explainable drivers. Applied as a full behavioral model, the same digital-twin representation opens up much more, including relationship and sub-tier exposure, cohort movement, and drift that surfaces before conventional supplier metrics react, giving DLA earlier and more connected supplier-risk insight.

6.3 Supplier Risk Detection Scenarios

Scenario	Traditional Visibility	BEI Signal
Supplier sourcing shift to adversary nation	May remain invisible until annual audit or shipment review	The geographic risk zone drifts while the performance zone stays stable
Hidden supplier concentration	Suppliers appear independent when reviewed separately	Network cohort analysis flags correlated supplier behavior
NSN transitioning to stockout	Visible only when a stockout occurs	Inventory position zone accelerates toward depletion regime
Counterfeit part entry	Often detected after a field failure or inspection issue	Supplier x NSN failure divergence and component relationship drift
Supplier financial distress	Detected at the next quarterly or external review	Performance and financial zones drift concurrently

Table 4 - Supplier risk detection scenarios.

The supplier use case should be validated as a risk-intelligence capability, not merely as a replacement for existing supplier scores. Current systems remain important. BEI adds continuous behavioral direction, relationship-level visibility, calibrated failure probability, timing through survival modeling, and explanations that can be reviewed before risk becomes delivery failure or readiness impact.

7. Use Case 3 - Logistics Network Early Warning and Mission Readiness Risk

DLA manages an interconnected network of NSNs, depots, suppliers, routes, and platforms in which readiness risk can build quietly across many entities at once. A part's availability can erode through a chain of small shifts, a supplier slipping, a depot buffer thinning, a route congesting, or a shared sub-tier wobbling, none of which crosses a threshold on its own, until the result appears as a stockout or a mission-readiness gap. This is the risk Use Case 3 targets: slow, directional, relationship-level change that no single system surfaces.

The mechanism for this already exists in the platform. The graph-based risk-propagation and cross-entity correlation components operate on logistics entities, linking NSNs, depots, suppliers, routes, and sub-tier dependencies into one network and tracing how risk moves through it. What remains to be quantified on DLA data is how early and how reliably that network view warns of readiness risk.

The underlying detection principle, finding slow, hidden, directional change that stays within conventional thresholds, has been validated most rigorously in a cybersecurity proof of concept. There, the same behavioral approach identified four long-duration intrusion campaigns that threshold and signature-based tools missed, while no conventional detector caught more than one. The two domains differ in data, but they share the structure the framework is built to detect, which is why that result is meaningful evidence that the same mechanism can surface slow, networked, sub-threshold risk in logistics. This is offered as a transferable detection principle, that slow, persistent, sub-threshold signals are discoverable, and as a hypothesis to validate on DLA logistics data, not as a logistics performance result.

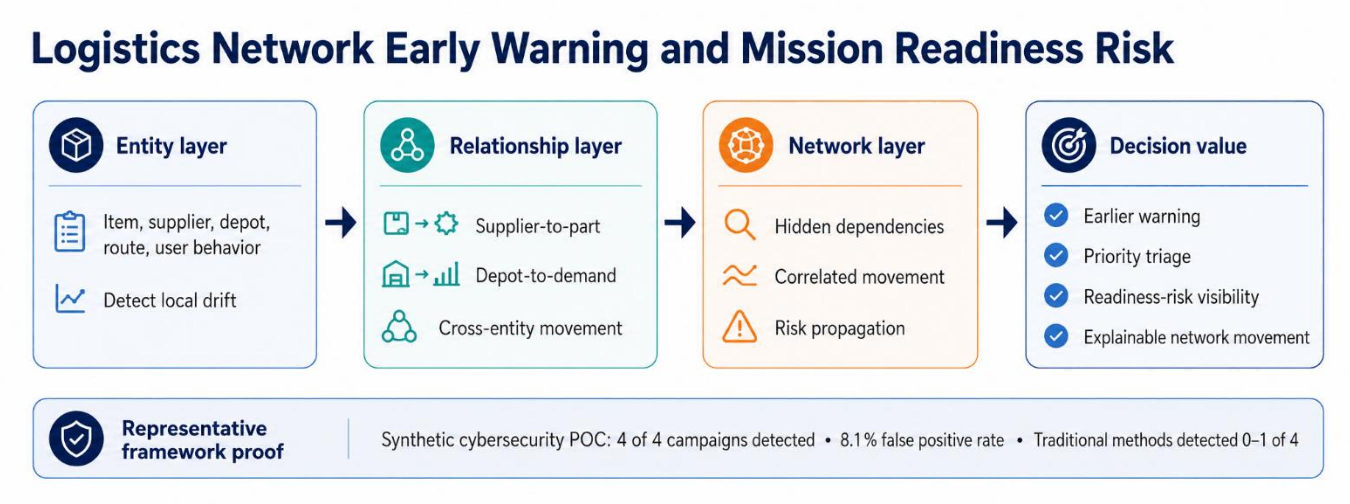

Figure 8 - Logistics Network Early Warning and Mission Readiness Risk

This is where the three-layer architecture earns its place. Entity profiles flag drift in a single item, supplier, or depot. Relationship profiles catch pairwise movement that a single-entity review misses, such as two suppliers drifting together. Network intelligence surfaces what only the graph shows: hidden dependencies, correlated movement, risk propagation, and coordinated behavior. For DLA, this is the basis of an early-warning lane for replacement-part readiness, hidden supplier concentration, depot-specific stockout exposure, and route-disruption sensitivity, with software bill of materials (SBOM) drift and broader mission-facing logistics risk added as the framework extends to richer data. How early and how reliably it warns on each of these is what the validation pilot would measure.

7.1 Approach Proof and Transfer Logic

Proof Area	Source Detail	DLA Transfer Logic
Cyber proof of concept population	250 users across employees, contractors, and service accounts	Demonstrates entity profiling at the population scale
Observation period	485 days from January 2025 to May 2026	Shows long-duration temporal drift tracking
Event volume	Approximately 14 million events across 5 log sources	Demonstrates capacity to process large behavioral telemetry

Proof Area	Source Detail	DLA Transfer Logic
Attack campaigns	4 long-duration campaigns: insider threat, slow APT C2, Volt Typhoon LOTL, Salt Typhoon telecom	Shows ability to detect slow, sophisticated movement
Detection result	4 of 4 campaigns detected at 8.1% false positive rate	Shows directional scoring can detect hidden behavior
Traditional comparison	Z-score detected 1 of 4, LOF / Isolation Forest / OCSVM detected none	Shows magnitude-only methods missed directional movement
Triage value	Zone-level directional intelligence identified which behavioral dimension changed	Shows outputs can be explained, not just scored
Prior-art discipline	Novelty is composition: semantic representation, direction, cohort, relationship, network, and multi-phase scoring	Protects claim credibility and avoids overstating any single method

Table 5 - Cybersecurity proof of concept evidence (synthetic data).

Note: *these are cybersecurity proof-of-concept figures that establish the detection principle; they are not a logistics performance claim. The 8.1% false-positive operating point was selected with knowledge of the ground truth and is a reporting benchmark, not a deployable threshold.*

7.2 DLA Mission Patterns Supported by BEI

- Replacement-part readiness exposure: part demand, supplier capacity, inventory position, and platform context drift toward risk before visible shortage.
- Hidden common dependency: nominally independent suppliers show correlated behavior because they share a sub-tier source, geography, process, or manufacturer.
- Route or location sensitivity: NSN x depot or supplier x location behavior changes in ways that indicate theater, route, or depot-specific exposure.
- SBOM or component drift: bill-of-materials composition changes without clear notification, creating counterfeit, compliance, or security risk.
- Demand diversion or fraud pattern evolution: demand composition, recipient location, or item relationship changes while aggregate demand volume appears normal.
- Cohort movement: multiple entities in the same peer group drift together, pointing to a shared cause such as market shock, geopolitical exposure, or supplier ecosystem stress.

This use case should not be positioned as a broad “detect everything” capability. It should be validated as an early-warning and triage layer for selected mission scenarios where existing DLA data can populate entities, relationships, cohorts, and networks. The strongest initial candidates are scenarios where individual metrics have historically remained acceptable until the risk was already operationally consequential.

8. Technical Evidence and Validation Boundaries

This section distinguishes proof of concept from operational proof. The current proof of concept is strong because it does more than describe a concept. It demonstrates that EDM / BEI can model DLA-relevant demand, supplier-risk, and network early-warning scenarios using controlled synthetic data, measurable baselines, calibrated outputs, explainable drivers, and defined validation boundaries. That level of evidence is sufficient to warrant a bounded pilot. The remaining question is not whether the concept is technically plausible. The remaining question is whether the same approach produces measurable mission value on DLA operational data.

The pilot should therefore be treated as the logical next validation step. It should test EDM / BEI in advisory, side-by-side mode against current baselines, SME review, and agreed success measures. The objective is to determine whether EDM / BEI improves decision usefulness in DLA's operating environment, including forecast calibration, procurement-plan accuracy, early-warning lead time, supplier-risk discrimination, relationship-level visibility, explainability, analyst usability, and integration with existing workflows. This framing protects claim credibility while giving DLA a clear, risk-managed basis for moving from proof-of-concept evidence to operational-data validation.

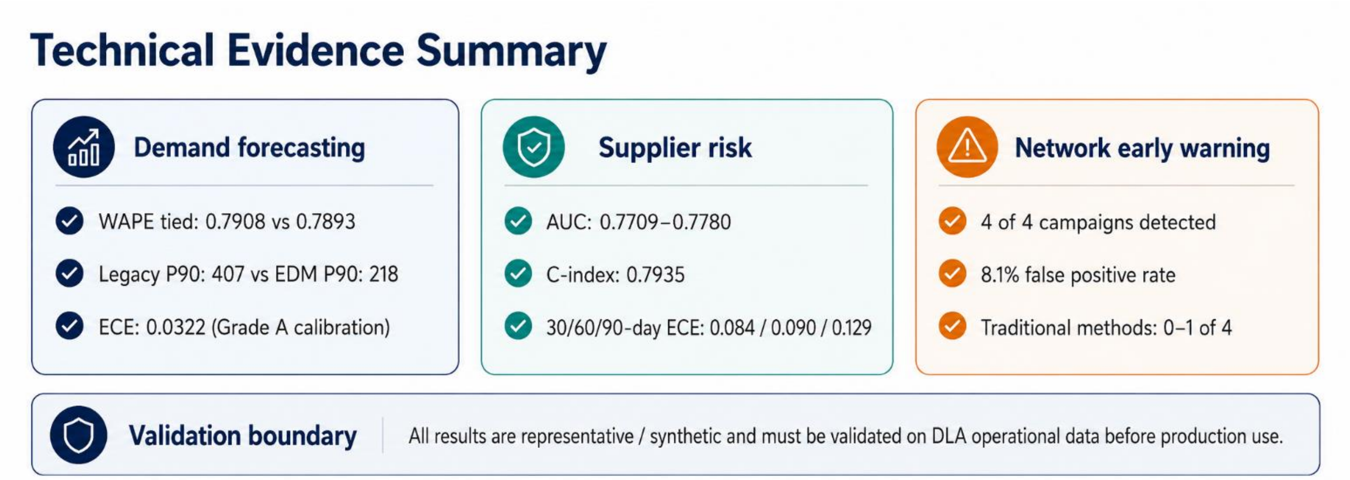

Figure 9 - Technical Evidence Summary

Evidence Category	What the Evidence Shows	How It Should Be Claimed
Demand forecasting	EDM creates calibrated, dynamic uncertainty bands and can materially alter procurement planning in representative synthetic scenarios	Claim as a strong basis for DLA data validation, not as operational superiority
Procurement value	Adapter Plate example shows 154 fewer P90 units and approximately $460K avoided over-procurement	Use as a representative synthetic proof of decision-value potential
Supplier risk	Behavioral modeling flags higher-risk suppliers, estimates time-to-failure, and explains its drivers	Claim as a qualitative proof of concept requiring real-data validation
Network early warning	BEI detected all 4 embedded cyber campaigns at 8.1% FP while traditional methods caught at most one	Claim as framework evidence and transfer logic, not a DLA logistics production result

Evidence Category	What the Evidence Shows	How It Should Be Claimed
Architecture	high-dimensional embeddings, cohort baselining, concept projection, relationship modeling, graph intelligence, and multi-phase scoring create a distinct composition	Claim composition-level differentiation, not novelty of every component
Deployment readiness	Containerized FastAPI, PostgreSQL / pgvector, HNSW index, SCD2 history, gold views, Docker, and air-gap option	Claim practical implementation ability for controlled validation
Data dependency	Existing DLA requisition, inventory, procurement, supplier, quality, COO, SBOM, and financial feeds can populate the model	Validate data access, quality, entity resolution, and governance before pilot expansion

Table 6 - Consolidated evidence and how it should be claimed.

8.1 Technical Evidence Summary

- EDM demand forecasting uses a two-stage hurdle LightGBM model with occurrence classification and quantile regressors for P10 through P90.
- Demand entities are represented in a high-dimensional semantic vector space using demand, supply, inventory, volatility, and context signals.
- Demand Envelope construction captures identity, time series, demand decomposition, supply network, inventory position, financial profile, operational context, and embeddings.
- EDM drift uses velocity, acceleration, mean-step drift, consistency, concept alignment, bounded alpha, and dynamic band multipliers of approximately 0.85x to 1.6x.
- Demand safeguards include +/-30% alpha cap, weak-alignment zeroing, inconsistency dampening, and protection for exercise-sensitive rising demand.
- Supplier risk uses entity profiling, zone decomposition, relationship analysis, temporal trajectory, regime detection, XGBoost classification, Random Survival Forest timing, SHAP explanation, and monotone constraints.
- BEI framework uses nine dimensions, three analytical layers, self-baseline and cohort-baseline comparisons, concept naming, CUSUM-style slow-drift accumulation, relationship profiles, and multi-phase fusion.
- The cybersecurity proof of concept remains important because it demonstrates the framework's ability to detect directional movement that traditional magnitude-based methods missed.

9. Recommended Pilot Validation Path

The next step should be advisory, side-by-side validation on DLA operational data. The purpose is not to displace current systems in one move. The purpose is to determine whether EDM/BEI provides measurable decision value beyond current baselines for selected logistics decisions. That path protects DLA's mission, preserves the discipline, and creates an evidence-based basis for pilot expansion if results warrant it.

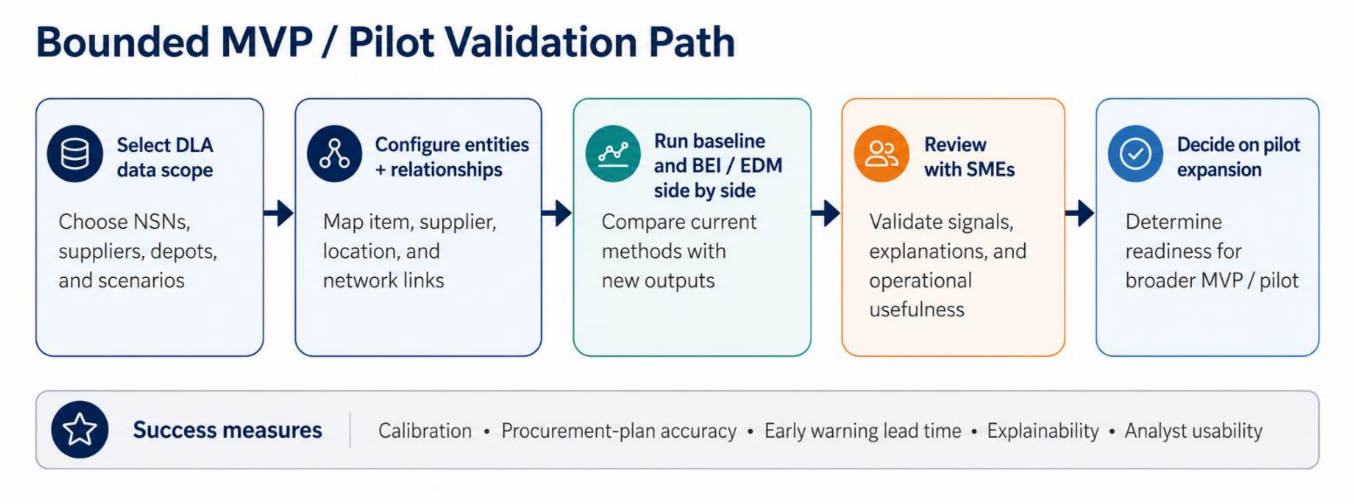

Figure 10 - Bounded Pilot Validation Path

The pilot should run in parallel with existing planning and monitoring processes. It should not feed automated procurement or supplier action until results are validated by subject matter experts (SMEs) and decision owners. The initial scope should be narrow enough to execute quickly and broad enough to test the three use cases, in priority order, meaningfully.

pilot Element	Recommended Scope	Success Measures
Demand forecasting	Selected NSNs, depots, high-value intermittent items, exercise-sensitive items, and sole-source critical items	Calibration, pinball loss, MASE, procurement-plan accuracy, dynamic band usefulness, and early warning
Supplier risk / SCRM	Selected suppliers, commodity groups, NSNs, COO / SBOM records, quality, and delivery history	Risk discrimination, survival timing, explainability quality, relationship coverage, false-positive review, SME usability
Network early warning	Supplier x NSN, NSN x Location, Supplier x Country, Supplier x Supplier, BOM relationships	Relationship drift detection, hidden dependency identification, risk propagation, triage usefulness
Data readiness	Requisitions, orders, inventory, procurement, delivery, quality, SPRS, Product Data Reporting and Evaluation Program (PDREP), Government-Industry Data Exchange Program (GIDEP), SAM.gov, PIEE / WAWF, DIBBS, COO, SBOM, financial feeds	Completeness, data lineage, entity resolution, timeliness, access control, security posture
Governance	SME review board, model audit, claim discipline, threshold tuning, operational review cadence	Confidence in outputs, explainability, operational acceptance, and bounded decision use

pilot Element	Recommended Scope	Success Measures
Pilot expansion decision	Evidence review after side-by-side validation window	Proceed, refine, expand, or stop based on measured mission value

Table 7 - Recommended pilot scope and success measures.

9.1 Suggested Pilot Operating Model

- Operate in read-only advisory mode during the initial validation window.
- Run existing baselines and EDM / BEI outputs side by side so DLA can compare signal quality without disrupting operations.
- Use historical back-testing and forward-looking shadow mode to evaluate both retrospective and live decision usefulness.
- Require SME review of explanations before treating a signal as operationally meaningful.
- Publish validation results with a clear distinction between baseline performance, EDM / BEI performance, synthetic carryover assumptions, and real-data findings.
- Maintain model and data governance from the start, including provenance, security, access control, calibration monitoring, and documentation of limitations.

10. Implementation Considerations

The implementation details for EDM / BEI will be addressed in a follow-on implementation strategy paper. This innovation concept paper establishes the mission need, technical proof of concept, validation boundaries, and recommended pilot path. The next paper will translate that foundation into an executable implementation plan that DLA can use to evaluate scope, schedule, data requirements, security posture, governance, success measures, level of effort, and the appropriate acquisition pathway. It will also name the target integration surface, including the Enterprise Business System (EBS), the demand-planning system, and the DLMS transaction layer, and the closed-loop path by which a validated planning band re-enters the system of record rather than remaining a side-by-side view.

The follow-on paper will define how the recommended pilot will be structured and executed, including the initial use-case scope, data-access assumptions, system integration approach, deployment options, security and authorization considerations, model governance, SME review process, validation scorecard, delivery timeline, and transition criteria. It will also support acquisition planning by outlining how the pilot could be pursued.

11. Conclusion

DLA's highest-value logistics decisions increasingly depend on detecting movement before failure is visible. In a contested logistics environment, adversarial exploitation and disruption can affect demand, supplier behavior, sourcing exposure, route vulnerability, sub-tier dependency, cyber-enabled continuity, inventory posture, and correlated network risk before conventional thresholds are breached. This is the decision environment EDM / BEI is designed to address.

The consolidated evidence is strong, but it should remain bounded. EDM has demonstrated a technically specific path for drift-augmented, uncertainty-aware demand forecasting. BEI has shown supplier-risk modeling that flags higher-risk suppliers, estimates time-to-failure, and explains its drivers. The broader BEI framework has demonstrated directional early-warning detection in a proof of concept where traditional magnitude-based methods did not surface the embedded campaigns. These results should not be overstated as DLA production proof. They should be treated as a compelling proof of concept that justifies operational-data validation.

The practical recommendation is clear. DLA should evaluate EDM / BEI through a bounded pilot that compares existing baselines and behavioral-intelligence outputs side by side across selected demand, supplier-risk, and network early-warning scenarios. The validation should measure the mission decision value that matters: calibrated uncertainty, procurement-plan accuracy, early-warning lead time, supplier-risk discrimination, survival timing, explainable drivers, relationship-level risk detection, analyst usability, and workflow fit.

Once validated on DLA operational data, EDM / BEI will not simply add another model or dashboard to the logistics environment. It will provide a decision-intelligence layer that helps DLA understand where demand, supplier behavior, and network risk are moving before the operational consequence becomes visible. That capability directly supports mission assurance, operational resiliency, supply-chain risk mitigation, and decision advantage in a contested logistics environment.

Based on the assessment in this paper, we recommend that DLA begin with Use Case 1 - Demand Forecasting and Inventory Planning as the initial pilot focus. This use case provides the clearest path to validate EDM / BEI against existing demand, supply, inventory, and procurement-planning data while producing measurable outputs tied to forecast quality, uncertainty management, procurement-plan accuracy, early warning, and decision advantage. The pilot scope, assumptions, data-access requirements, schedule, level of effort, and success measures should be refined jointly with DLA and confirmed through mutual agreement before execution. Once aligned, 22nd Century Technologies is prepared to work with DLA through the appropriate contractual vehicle and acquisition steps needed to execute the recommended approach.

Appendix A: Glossary of Key Terms

EDM (Entity Digital Model) Vector Intelligence. an approach that rebuilds each item, supplier, depot, and route as a time-aware digital twin and represents its state as a high-dimensional vector, so it can be compared over time and against peers.

BEI (Behavioral Entity Intelligence). the cross-domain early-warning framework that analyzes entities, their relationships, and the network over time to surface slow, directional change.

NSN (National Stock Number). the standard identifier for a supply item; the first four digits are the Federal Supply Class.

DODAAC. Department of Defense Activity Address Code; identifies the activity or location associated with a requisition.

P10 / P50 / P90. points on the forecast range: the P90 is the level demand is not expected to exceed about 90 percent of the time, used here as the planning (buy) quantile; the P50 is the midpoint.

Calibration (ECE, Brier score). measures of whether predicted probabilities match real frequencies; lower values mean the forecast ranges can be trusted.

Hurdle model. a two-stage forecast that first predicts whether demand will occur at all, then how much, which suits intermittent demand.

Drift / CUSUM. drift is gradual, directional change in behavior; CUSUM is a change-detection method that accumulates small shifts until they become significant.

Cohort. a peer group of similar items or suppliers; behavior is judged relative to the cohort, and new items borrow from their nearest behavioral twins.

Survival model. a method that estimates the time until an event such as a supplier disruption and how that risk develops over 30, 60, and 90 days.

Embedding. a numerical vector that captures the meaning of serialized entity state; here it generates most of the engineered features (zone, drift, and cohort features).

Criticality code. an item designation (for example C or B) reflecting mission importance, used to set how protective the planning level should be.

SHAP. a method that explains a model's output by attributing it to individual input features, so each forecast or risk score is traceable.

MASE (mean absolute scaled error). a scale-free forecast-accuracy measure suited to intermittent demand; below 1 means better than a naive baseline.

Pinball loss. the standard scoring rule for quantile (range) forecasts such as the P90; lower is better.

SBOM (software bill of materials). a structured list of the components in a software product, used to track supply-chain and component risk.